# India's Imports: Trade Insights from Asian Countries Using Python (2023–2024)

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## 1. Project Overview

This project analyzes India's import trade data from Asian countries using Python. The dataset includes transaction-level information such as country, commodity, quantity, unit of measurement, and import values in INR and USD.

The primary goal is to transform raw trade data into actionable insights through structured data cleaning, statistical analysis, and visualization. The project helps uncover trade patterns, identify key contributors, and understand how import behavior varies across countries, commodities, and time.

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## 2. Dataset Information

- **Domain:** Finance (International Trade Analytics)
- **Source:** Indian Open Data Portal
- **Region:** Asia
- **Available Timeline:** 2015 – June 2025
- **Selected Period:** 2023 – 2024

The dataset includes:

- Transaction date
- Country and sub-region
- Commodity details (HS Code + name)
- Unit of quantity
- Quantity imported
- Import value in INR and USD

The dataset was filtered to recent years (2023–2024) to ensure **relevance and computational efficiency**.

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## 3. Problem Statement

Trade datasets are large and complex, making it difficult to derive meaningful insights without proper analysis. Understanding import patterns is essential for:

- Identifying key trading partners
- Tracking high-value commodities
- Supporting strategic decision-making

This project aims to analyze import data to uncover **patterns, trends, and dependencies** in India's trade with Asian countries.

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**4. Objectives**

- Understand dataset structure and variables
- Clean and preprocess raw data
- Handle missing values and inconsistencies
- Perform exploratory and statistical analysis
- Visualize trends and relationships
- Generate business-relevant insights

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**5. Stage 1 – Data Understanding & Initial Exploration**

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Initial exploration was performed to understand dataset structure and quality.

**Key Steps:**

- Loaded dataset from GitHub
- Inspected data using head(), tail()
- Checked shape and column details
- Analyzed data types and summary statistics
- Identified missing values and duplicates

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**Insights from Stage 1:**

- Dataset contains **large-scale transactional data**, suitable for trend analysis
- Numerical columns were incorrectly stored as **strings**, requiring conversion
- Minor missing values were present in **Unit of Quantity**
- No duplicate records → data reliability is high

## 6. Stage 2 – Data Cleaning & Preprocessing

This stage focused on improving the quality of the dataset by handling inconsistencies, correcting data types, and preparing the data for analysis. Proper data cleaning ensures that the dataset is accurate, reliable, and suitable for generating meaningful insights.

### 6.1 Removing Redundant Columns

The dataset contained both **Region Name** and **Sub-Region Name** columns, along with their respective codes. Since the **Sub-Region column already provides more specific and detailed geographic classification**, the Region-related columns were considered redundant.

- Dropped columns:
    - Region Name
    - Region Code

Removing these columns helps:

- Reduce unnecessary data duplication
- Simplify the dataset structure
- Improve clarity and focus during analysis

### 6.2 Handling Missing Values

- Identified missing values in categorical and numerical columns
- Applied:
    - **Mode imputation** for categorical data
    - **Median imputation** for numerical data

### 6.3 Fixing Data Types

The numerical columns in the dataset, such as **quantity** and **import values (INR and USD)**, were initially stored as text (object data type). This can lead to incorrect calculations and errors during analysis.

- Converted the following columns to numeric format:
    - Quantity of commodity
    - Value of commodity in INR
    - Value of commodity in USD

👉 This step helps to:

- Enable accurate statistical calculations (mean, median, etc.)
- Ensure proper visualization and analysis
- Detect and handle invalid or inconsistent values (converted to NaN)

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**6.4 Cleaning Unit of Quantity**

- Detected inconsistent values due to parsing errors
- Defined valid units (Kgs, Nos, Mtr, etc.)
- Replaced invalid entries with NaN
- Imputed using mode

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**6.5 Final Validation**

- Verified no missing values
- Confirmed correct data types
- Ensured dataset consistency

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**Insights from Data Cleaning:**

- Real-world datasets often contain **formatting issues and inconsistencies**
- Unit column required careful cleaning → critical for accurate analysis
- Median imputation was effective due to **skewed distribution of values**

## 7. Stage 3 – EDA, Statistical Analysis & Visualization

### 7.1 Statistical Analysis

**Key Measures:**

- Mean, Median, Mode
- Variance & Standard Deviation
- Skewness & Kurtosis

**Insights:**

- Mean > Median → confirms **right-skewed distribution**
- High standard deviation → large variability in import values
- Skewness indicates presence of **extreme high-value transactions**
- Trade data is **not normally distributed**

### 7.2 Time-Based Analysis

- Extracted Year and Month from Date

**Insights:**

- Import activity varies over time
- Indicates **seasonality and demand fluctuations**
- Useful for forecasting and planning

### 7.3 Univariate Analysis

**Key Findings:**

- Most transactions involve **small quantities and values**
- Distribution is heavily skewed
- Top countries dominate transaction count
- Few commodities appear frequently
- Units like **Kgs and Nos dominate**, indicating physical goods trade

### 7.4 Bivariate Analysis

**Key Relationships Identified:**

- **Country vs Import Value:**
  Few countries contribute the majority of import value
- **Commodity vs Value:**
  Certain commodities dominate trade → high dependency
- **Quantity vs Value:**
  Positive relationship, but not linear → price variation exists
- **Time vs Value:**
  Import values fluctuate → indicates economic or seasonal impact

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**Key Insights:**

- Trade is **highly concentrated**
- Not all commodities contribute equally
- High-value transactions drive overall trade

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### 7.5 Multivariate Analysis

**Insights from combined analysis:**

- Strong correlation between INR and USD values
- Quantity and value show moderate relationship
- Certain countries dominate specific commodities
- Trade patterns vary across combinations of:
  - Country
  - Commodity
  - Time

## 8. Stage 4 – Insights, Recommendations & Conclusion

### 8.1 Key Insights

- Import trade is **highly concentrated among a few countries**
- A limited number of commodities contribute to most of the value
- Data is **heavily skewed with outliers**
- Majority of transactions are low-value
- Trade patterns vary over time
- Measurement units indicate **physical goods dominance**

### 8.2 Business Insights (Important Upgrade)

- India shows **dependency on specific countries**, indicating risk
- High-value commodities are critical to supply chain
- Trade imbalance suggests **opportunity for diversification**
- Seasonal variations indicate planning opportunities

### 8.3 Recommendations

- Diversify import sources to reduce dependency
- Focus on high-value commodities for optimization
- Use time trends for demand forecasting
- Monitor large transactions for financial control
- Improve procurement strategies using data insights

### 8.4 Future Enhancements

- Add recent data for better accuracy
- Build dashboards for real-time insights
- Apply machine learning for forecasting
- Include macroeconomic factors

### 8.5 Conclusion

This project successfully transformed raw import trade data into meaningful insights using Python. Through data cleaning, statistical analysis, and visualization, key trade patterns and dependencies were identified.

The findings highlight concentration of trade, skewed data distribution, and the importance of certain commodities and countries. These insights can support better strategic decision-making in international trade.

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